Intentions Matter: The Role of Purpose in Excessive Social Media Usage and Smartphone Addiction

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Abstract— **Social media offer a wide range of advantages, but excessive usage can result in negative, psychological and behavioral consequences, including addiction. This study aims to explore the role of intentions of excessive social media usage in smartphone addiction. The analysis included data from 483 respondents who answered an online survey. The results showed that intentions of excessive social media usage can be grouped into four categories: social motivations, mode modification, productive reasons, and peer pressure. The results of multiple linear regression analysis revealed that the using social media excessively for mode modification and for peer pressure were significant predictors of smartphone addiction. Additionally, multivariate analysis of variance (MANOVA) indicated significant age groups differences in intentions of excessive social media usage for peer pressure and for productive reason. Emerging adults used social media excessively for peer pressure and productive reasons compared to adults. Conversely, a significant gender-related differences in intentions of excessive social media usage for mode modification and for social motivation. Females used social media excessively for mode modification and for social motivation compared to male. These findings enhance our comprehension of the factors impacting smartphone addiction and emphasize the role of intentions of excessive social media usage in its development, overcoming the limitation of the literature on screentime, which has paid limited focus to the purpose of use.**

Keywords—smartphone addiction, digital wellbeing, intention, excessive social media usage, social motivation, peer pressure, mood modification

I. Introduction

Nowadays, smartphones have become an indispensable tool in our daily lives. They are designed to provide instant access to online platforms and applications. With smartphones, users can send messages and e-mails, listen to music, take pictures and videos, play games, and use social networking [1]. The number of smartphone users increased by 73.88% from 2016 to 2021 [2]. Social media applications are the most widely used [3] and fastest growing among various mobile applications, and according to the data, the world top three most downloaded applications are TikTok, Instagram and WhatsApp [4]. Excessive use of social media applications on mobile phones can lead to smartphone addiction.

Recent research into smartphone addiction has attracted considerable attention to explored the risks of excessive and uncontrolled use of smartphones. Smartphone addiction and internet addiction are not yet recognized as a formal clinical disorder in the latest version of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) [5]. Researchers have found similarities between smartphone usage patterns and behavioral addiction symptoms include withdrawal, increased tolerance, interpersonal conflicts, mood modification and salience [6]–[8]. Several terms were used to describe smartphone addiction "such as “problematic mobile phone use” [9], “cell phone addiction” [10] and “smartphone addiction” [11]. Smartphone addiction can be understood in the broader context of technological addictions. Technological addictions, as defined by Griffiths [12], involve non-chemical behavioral addictions arising from human-machine interactions.

Excessive social media use is one common symptom of smartphone addiction. We argue that it is essential to consider individuals intentions of excessive social media use, when evaluating the impact of that usage on smartphone addiction. Besides smartphone usage, research has shown that social media usage in particular can be addictive [13]–[15]. Several factors contribute to this addictive relationship including individual traits, social influences, environmental factors, and the role of design. Additionally, the design of social media platforms may encourage excessive social media usage [16]. Due to their business model, these platforms are typically designed to prolong usage, through some features. One of the feature is infinite scrolling which is provide a continually refreshing list of content as users scroll down [17]. Noë et al. [18] study revealed that infinite scroll feature correlates with problematic smartphone use. Another feature is constant notifications which act as reward mechanisms. A simple notification, like a friend liking a post, can lead to a 15-minute social media binge during the workday [19]. These design features can lead to social media addiction by generating a loop of involvement that is difficult for users to quit.

Several social factors contribute to the development of addictive social media usage. Among these factors, peer pressure is a significant influence, as friends may expect immediate responses and comments on their posts [20]. The

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desire to fit in and maintain social connections can lead individuals to spend excessive amounts of time on their smartphones, facilitating smartphone addiction. A study found that peer pressure significantly predicted mobile social media addiction in adolescents [20]. Addictive social media tendencies have also been associated with the human need for social support [21]. Individuals use these platforms to seek validation and empathy from online connections by sharing thoughts, experiences, and emotions. Furthermore, some may use social media excessively to escape negative moods and realities, finding relief through virtual relationships and alternate worlds [21]. Understanding the diverse social factors that sway people to overuse social media can inform the design of platforms and interventions to help mitigate these risks. Recognizing the role of social pressure, belonging, validation-seeking, and escapism is key to addressing the roots of addictive social media engagement and, in turn, smartphone addiction.

In addition to the factors which are related to maladaptive behaviors, excessive use of social media can also be attributed to positive outcome. For example, usage can be to stay updated on peers' academic activities while facilitating building professional networks and creating professional online profiles [21, 22]. Platforms, such as Twitter, Facebook, and WhatsApp, have proven to be valuable tools for social science academics, allowing them to effectively engage with their audience and leverage interactive features for research activities and the dissemination of research findings [24]. Research in [25] suggested that people with high education tend to be more active users of social media platforms and attributed this behavior to their increased need for information. However, recent findings highlighted the need for teachers and educators to be aware of the potential psychological effects of excessive social media use, given the considerable rise in social media usage among students, influenced by globalized communication and post-pandemic changes in education [26].

This paper argues that intentions behind excessive social media usage helps a nuanced understanding of the relationship between the general notion of screentime and smartphone addiction. Our research questions (RQs) are as follows:

RQ1: Do differences in intentions for excessive social media usage exist among age groups?

RQ2: Are there gender-related differences in intentions for excessive social media usage?

RQ3: Can intentions for excessive social media usage predict smartphone addiction?

II. Dataset

The data were collected using a mobile application that is available on Google Play. The app enables users to track their screen time and phone usage to promote smartphone awareness. When initiating the app usage, users were invited to participate voluntarily, grant their consent, and complete survey questions. Moreover, users of the app acknowledged and agreed to the privacy policy of the company that owns the app, which described that data are collected for research purposes, and assured anonymity of the data. A total of 640 users from different countries responded to the survey between November 2022 and March 2023. The largest percentage of respondents were from the United States (41%), followed by the United Kingdom (22%), France (9%), and Germany, Canada, and India (5% each). Australia and the Netherlands accounted for approximately 2% - 4% of the respondents each, while the remaining countries constituted 8% of the total participants.

A total of 157 respondents were excluded from the analysis for one or more of the following reasons: failure to answer the two attention checks questions, completing the survey in very short time, participants aged less than 15, participants who did not answer the questions relevant to this study and participants providing contradictory responses. Additionally, the small number of participants who reported their gender as "other" or had missing gender information were excluded in the analysis of the present work. The final analysis was conducted on a sample size of 483 participants.

III. Measures

A. Demographics questions

Participants were requested to provide information regarding their age and gender.

B. Smartphone addiction short scale

The smartphone addiction short-scale version (SAS-SV) proposed in [27] was used. It comprises 10 items on a six-point Likert scale, ranging from 1 ("strongly disagree") to 6 ("strongly agree"). The summation of these responses results in a total SAS-SV score ranging from 10 to 60, with a higher score indicating a higher risk of smartphone addiction. The reliability of the scale was assessed using Cronbach's alpha coefficient $\alpha = .772$ indicating good internal reliability [28].

C. Intentions of excessive social media usage

The intentions of excessive social media usage were assessed using eleven statements. Participants were asked about their reasons for excessive social media usage, including whether this was due to: 1) essential purposes such as work or study, 2) the pressure of my peers to be online and to react to their posts and comments, 3) change my mood, 4) bond with others, 5) get support from others, 6) provide support to others, 7) search opportunities, 8) pass time when I have nothing else to do, 9) my desire to make a difference or make an impact through my posts, 10) the way social media are designed that makes me more attached to them and 11) my personal choice and I have control over this pattern of use. A six-point Likert scale was used to rate these statements, ranging from 1 ("never") to 6 ("very frequently"). The removal of item 11 resulted in an increase in internal consistency as indicated by Cronbach's alpha from $\alpha = .745$ to $\alpha = .763$. Moreover, item 11 is not a reason but rather a declaration of consciousness of choice which further justifies its removal.

IV. Data Analysis

The data were imported into the statistical software JASP [29] for analysis. Descriptive statistics of frequency and percentages were obtained for age groups and gender. The independent sample t-test was used to examine the potential

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differences in smartphone addiction scores between gender and age groups. The data was checked for independent sample t-test assumptions. Boxplot analysis did not reveal any outliers. The skewness, kurtosis, and Q-Q plot indicated that the age groups and gender followed approximately a normal distribution, and the homogeneity assumption was confirmed using Levene's test.

To determine whether the intentions of excessive social media usage could be categorized into meaningful factors, Principal Component Analysis (PCA) was employed. Kaiser–Meyer–Olkin (KMO) and Bartlett tests were used to determine the suitability of the data for PCA. The KMO score measure of sampling adequacy was well above the recommended level of 0.50 [30]. The Bartlett test of sphericity was significant [31], indicating a sufficient level of intercorrelation between the 10 statements. Therefore, PCA was employed. Eigenvalues greater than 1.0 were used to determine the number of factors to retain in the PCA. Varimax rotation was conducted on the intention of excessive social media usage related items. Only variables with loadings of 0.5 or greater were considered as part of the suggested components [30]. Cronbach's alpha was used to assess the internal consistency of the items that make up the suggested subscales. Subscale scores were calculated for each participant based on the results of the PCA. Individual scores were added to each subscale that contained variables loaded at least 0.5 on the identified components and then divided by the number of items in each subscale.

MANOVA analysis was used to examine differences in the categories of intentions of excessive social media usage factors among gender and age groups. The normality and homogeneity assumptions were validated. Mahalanobis distances were computed to detect multivariate outliers, and no outliers were found using the criterion of a Mahalanobis distance of $p < .001$.

Furthermore, Pearson's correlation coefficients were used to examine the relationship between smartphone addiction scores and the intentions of excessive social media usage. Multiple regression was employed to investigate the influence of intentions factors of excessive social media usage as predictor variables on smartphone addiction served as the outcome variable. These variables were checked for multiple regression assumptions. The variance inflation factor (VIF) was employed to assess the multicollinearity of the independent variables, and it indicated an acceptable degree of collinearity (all VIFs < 2). Cook's distance analysis showed the residuals were normally distributed and no individual data points had undue influence on the regression model.

The final data for this research is published on the Open Science Framework repository at the following URL: https://osf.io/pjuzk/?view_only=18186bd3ff3a4c9fa644b032b41941c2 .

V. Results

A total of 483 participants were included in the analysis, consisting of 52% females and 48% males. Most of the respondents were adults (aged 25 to 64 years), while 46% were emerging adults (aged 15 to 24 years). The results of the independent sample t-test, as presented in Table I, indicated a statistically significant difference in smartphone addiction between males and females with a small effect size (t (482) = -2.711, p =.007; Cohens' d = -.247). Smartphone addiction score was higher in females (M = 37.85, SD = 8.67) compared to males (M = 35.7, SD = 8.73). Additionally, smartphone addiction was significantly higher in emerging adults (M = 38.4, SD = 8.69) compared to the adults group (M = 35.44, SD =8.60) but with a small effect size (t (482) = 3.761, p <.001; Cohens' d = .343).

TABLE I: INDEPENDENT SAMPLE T-TESR RESULTS OF SMARTPHONE ADDICTION WITH GENDER AND AGE GROUPS DIFFERENCES

Variable		N (%)	M (SD)	*t*(482)	*P*	Cohens' *d*
Gender	Male	234 (48%)	35.7 (8.73)	-2.711	.007	-.247
	Female	249 (52%)	37.85 (8.67)			
Age group	Emerging adults	223 (46%)	38.4 (8.69)	3.761	<.001	.343
	Adults	260 (54%)	35.44 (8.60)			

The PCA result indicated that the first factor accounts for 33.7% of the variance, the second accounts for 13.3%, and the third explains 10.6% of the variance. All the three-factor accounted for 57.6% of the total variance. Out of ten statements, nine were highly correlated with each other. Four items loaded into the first factor and were labeled "Social motivations". This factor encompassed the reasons of excessive social media usage, which included: 1) to get support from others, 2) to provide support to others, 3) for bonding with others, and 4) to make a difference or have an impact through posts. Three items loaded into the second factor that was labeled, "mode modification". It encompassed the following reasons: 1) to change mood, 2) to pass time when nothing else to do, and 3) being more attached to social media due to their design. Two items loaded into the third factor that was labeled, "productive reasons". It consisted of 1) using social media for essential purposes such as work or study and 2) for searching opportunities. Peer pressure item was removed due to its low loading, therefore, it was investigated and considered separately in the analysis as an individual variable. The reviewed literature has demonstrated that peer pressure can independently predict mobile social media addiction [20]. Therefore, we analyzed peer pressure as a separate variable. These results are summarized in Table II.

The results of the MANOVA test using *Wilks'* λ showed a significant effects of age groups on intentions of excessive social media use (*Wilks'* λ = .973; F (1,481) = 3.286, p =.011). Follow-up univariate analyses were conducted to examine age groups differences in these intentions of excessive use. ANOVA revealed significant differences between age groups for peer pressure (F (1,481) = 4.632, p =.034, η^2 = .01) and productive reason (F (1,481) = 4.340, p =.038, η^2 = .009). The mean scores revealed that emerging adults (M = 2.18, SD = 1.38) experienced greater peer pressure for excessive social media use compared to adults (M = 1.93, SD = 1.20). Moreover, emerging adults (M = 3.08, SD = 1.20) reported a higher level of productive reasons for excessive social media use compared to adults (M = 2.86, SD = 1.18). While no significant differences were found between age groups with

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social motivation and mode modification as shown in Table III.

TABLE II: PRINCIPAL COMPONENT ANALYSIS OF ITEMS RELATED TO THE INTENTIONS OF EXCESSIVE SOCIAL MEDIA USAGE

Statements	PC1	PC2	PC3
Factor 1: Social motivations			
1. to get support from others	**.825**	.116	.051
2. to provide support to others	**.822**	.108	.070
3.or bonding with others	**.814**	.180	.047
4. to make a difference or have an impact through posts	**.594**	.092	.196
Factor 2: Mode modification			
1. to change mood	.133	**.732**	.253
2. to pass time when nothing else to do	.056	**.772**	-.094
3. being more attached to social media due to their design	.206	**.670**	.006
Factor 3: Productive reasons			
1. using social media for essential purposes such as work or study	.053	-.089	**.860**
2. for searching opportunities	.298	.199	**.626**
Eigenvalue	3.371	1.329	1.065
Proportion variance	33.7%	13.3%	10.6%
Cronbach's alpha, α	.800	.588	.421

The result of *Wilks' λ* showed there was a significant effect of gender on intentions of excessive social media use (*Wilks' λ* = .949; F (1,481) = 6.405, p <.001). Follow-up univariate analyses were conducted to examine gender-related differences in these four factors. Significant differences were found between gender for social motivation (F (1,481) = 14.698, p < .001, η^2 = .03) and mode modification (F (1,481) = 11.075, p < .001, η^2 = .023). The mean difference showed that females (M = 3.04, SD = 1.14) had a higher level of social motivation for excessive use of social media compared to males (M = 2.65, SD = 1.07). Similarly, for the mode modification factor, females (M = 4.70, SD = 1.06) had a higher level compared to males (M = 4.37, SD = 1.06). While no significant differences were found between gender with productive reasons and peer pressure as shown Table IV.

TABLE III: ANOVA RESULTS FOR INTENTIONS OF EXCESSIVE SOCIAL MEDIA USAGE BY AGE GROUPS

Factors	Age group	M (SD)	*F*	*p*	η²
Social motivation	Emerging adults	2.83 (1.08)	.089	.766	.000
	Adults	2.86 (1.16)			
Mood modification	Emerging adults	4.62 (1.01)	2.570	.110	.005
	Adults	4.47 (1.11)			
Productive reasons	Emerging adults	3.08 (1.10)	4.340	.038	.009
	Adults	2.86 (1.18)			
Peer Pressure	Emerging adults	2.18 (1.38)	4.632	.032	.010
	Adults	1.93 (1.20)			

TABLE IV: ANOVA RESULTS FOR INTENTIONS OF EXCESSIVE SOCIAL MEDIA USAGE BY GENDER GROUP

Factors	Gender	M (SD)	*F*	*p*	η²
Social motivation	Male	2.65 (1.07)	14.698	<.001	.030
	Female	3.04 (1.14)			
Mood modification	Male	4.33 (1.06)	11.075	<.001	.023
	Female	4.74 (0.93)			
Productive reasons	Male	3.00 (1.15)	.566	.452	.001
	Female	2.92 (1.14)			
Peer Pressure	Male	1.99 (1.26)	.864	.353	.002
	Female	2.1 (1.32)			

Pearson's correlation coefficients presented in Table V indicated significant positive correlation between the smartphone addiction score with social motivation (r = 0.244, p < .001) mode modification (r = 0.420, p < .001), productive reasons (r = 0.154, p < .001) and peer pressure (r = 0.256, p < .001).

TABLE V: PEARSON'S CORRELATION BETWEEN SMARTPHONE ADDICTION AND INTENTIONS OF EXCESSIVE SOCIAL MEDIA USAGE

Variables	1	2	3	4	5
1.Smartphone addiction	--				
2.Social motivations	.244***	--			
3.Mode modification	.392***	.343***	--		
4.Productive reasons	.155***	.343***	.172***	--	
5.Peer pressure	.256***	.391***	.204***	.205***	--

* p < .05, **p < .01, *** p < .001

The multiple regression model significantly predicted smartphone addiction, F (4, 478) = 28.319, p < .001, adj. R^2 = 0.19. The results indicated that the intention of using social media excessively for mode modification and peer pressure significantly predicted smartphone addiction. Excessive usage of social media for mood modification showed a significant positive effect on smartphone addiction (β = .334, p < .001). Similar findings were observed for intention of excessive social media usage due to peer pressure, with a significant positive effect on smartphone addiction (β = .158, p < .001) as presented in Table VI.

TABLE VI: MULTIPLE REGRESSION RESULTS FOR PREDICTING SMARTPHONE ADDICTION

Variables	*b*	*t*	*p*
Social motivations	.051	1.06	.290
Mood modification	.334	7.60	< .001***
Productive reasons	.048	1.08	.281
Peer pressure	.158	3.52	< .001***

* p < .05, ** p < .01, *** p < .001

VI. DISCUSSION

The purpose of this study is to investigate the relationship between the intentions of excessive social media use and smartphone addiction, considering potential gender and age differences. Furthermore, we seek to determine whether these intentions can predict smartphone addiction.

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Our results show that there are differences in smartphone addiction based on gender and age groups. Females and emerging adults showed higher level of smartphone addiction which is in line with other studies [32, 33]. Moreover, the peer pressure and productive reasons of intentions for excessive social media usage have a significant difference across the age groups. Specifically, emerging adults reported the highest mean of using social media for peer pressure compared to adults. This aligns with a study by Steinberg and Monahan, adolescents were found to be more susceptible to peer influence than adults [34]. Additionally, emerging adults have the highest mean of using social media for productive reasons compared to adults. Youth are typically at a stage in life where academic performance and career opportunities hold significant importance. Social media platforms beneficial students to enhance active learning [35]. Surveys suggest youth prefers online job search methods over traditional ones like newspapers [36]. As a result, emerging adults may feel compelled to use social media excessively for school-related tasks or to find opportunities to advance their careers. The developmental stage and goals of emerging adults appear to drive motivations for potentially excessive social media use.

The result showed that social motivation and mood modification factors have a significant difference a cross gender. Specifically, in terms of mode modification and social motivation, females scored higher means compared to males. This aligns with previous studies showing that females are more prone to social media addiction for the purpose of mood modification [37]. Similarly, the results of another study by Simsek, et al. [38] found that females achieved significantly higher mean scores for excessive social media usage for mood modification compared to males. Moreover, females scored higher than males in their excessive use of social media for social motivation. Previous research indicated that female users of social media are more motivated by social engagement, community interests, and personal gains than male users [39]. It is important to understand these differences in order to develop targeted interventions and strategies that can promote the use of social media and digital well-being both among men and women, in a tailored fashion.

On the other hand, our findings revealed a positive association between intentions of excessive social media usage (mode modification, social motivation, productive reasons and peer pressure) with smartphone addiction. Among these intentions, mode modification followed by peer pressure showed the highest correlation with smartphone addiction. This aligns with previous research showed that peer pressure and mode modification are positively correlated to smartphone addiction [20, 40]. Further supporting our correlation results, we found that mood modification and peer pressure significant predictors of smartphone addiction. Whereas social motivations and productive reasons, on the other hand, did not significant relationships with smartphone addiction. This result is consistent with a previous study that showed that social relationship and information seeking are not associated with smartphone addiction [40, 41].

Our findings indicated that the mode modification factor, encompassing intention of excessive social media use like mood modification, passing- time, and due to the influence of social media design exhibit higher levels of smartphone addiction. These results are consistent with previous studies showing the role of mode modification in smartphone addiction. Zhang et al, [40] found that mood regulation, pass time perceived, and enjoyment positively affect smartphone addiction. This finding can be explained in light of the self-regulation theory which suggests that people engage in certain behaviors to regulate their mood, emotions, and cognitive state [42]. For example, negative emotions such as sadness, boredom, or stress often prompt people to seek ways to improve their mood. Consequently, individuals turn to social media to regulate their moods or pass the time. Social media can offer people various activities and content to distract them from negative emotions and everyday situations. Scrolling through social media feeds, interaction with posts, or interacting with others may divert attention from negative thoughts and feelings and provide a temporary escape from reality. This relationship between negative emotions, mood regulation, and excessive social media usage may contribute to the development of smartphone addiction. Noë et al. [18] found that social media apps designed for smartphones are generally associated with smartphone addiction, reinforcing the significant association between intentions of excessive social media usage (specifically, more attachment to social media because of its design) and smartphone addiction.

The study revealed a significant association between excessive use of social media for peer pressure and smartphone addiction. Individuals who reported higher levels of peer pressure were more likely to exhibit addictive behaviors related to their smartphones. This finding aligns with previous research, which demonstrated that peer pressure to be online and participate in social media correlates with higher scores of smartphone addiction among individuals [24]. Another study found that peer pressure significantly predicted mobile social media addiction [20]. Social identity theory may provide an appropriate framework for understanding the influence of peer pressure on smartphone addiction. This theory states that individuals internalize the membership of groups to the extent that they become part of their self-concept, thus the identity depends on the membership of the relevant social groups [43]. Furthermore, the concept conformity occurs when an individual alters their own attitudes, beliefs, and behaviors to align with the prevailing norms of a group [44]. In other word, a person may be subjected to peer pressure if they strongly identify with a group (such as peers) and then demonstrate a tendency to embrace the group's views, attitudes, and behaviors in order to be accepted by the group as a whole. The constant presence and use of smartphones among peers can create a perceived need for individuals to be constantly connected, leading to smartphone addiction.

This study has some limitations. One limitation is the restriction to participants using Android phones only, as the data collection app was available only on Google Play store.

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However, there is no theoretical basis to think that the characteristics and behavior of iOS users will be different. We used self-report surveys for data collection, which might introduce reporting biases, a common issue in smartphone studies [45]. The cross-sectional design of our study prevents establishing causality between smartphone addiction and excessive social media use. To gain a more comprehensive understanding of this relationship, future research shall adopt longitudinal designs and incorporate objective measures to assess smartphone addiction and its association with intentions of excessive social media use.

VII. Conclusion

This research aimed to investigate the role of intention to excessive use of social media and its association with smartphone addiction. The findings revealed that using social media excessively for mode modification, and peer pressure were significant predictors of smartphone addiction. These results emphasize the importance of identifying specific excessive use and addressing the underlying factors that contribute to smartphone addiction. Understanding this relationship is crucial in developing effective interventions and strategies to promote healthier smartphone use and mitigate the negative consequences associated with excessive social media use and mode change.

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